

Supplementary information: Spatial association of
TLS programs and LAMP3⁺CCR7⁺ mregDC
states marks an immune-prognostic context in
ovarian cancer

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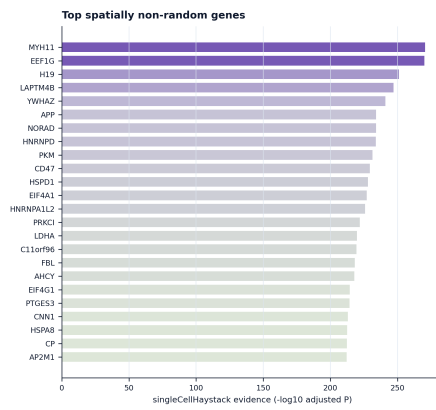
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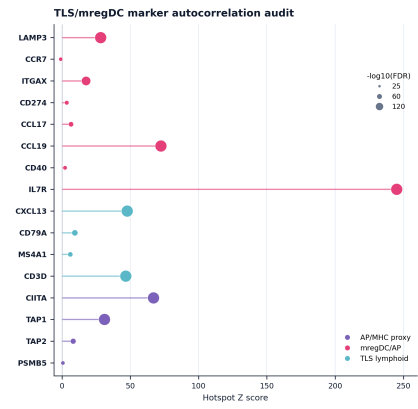
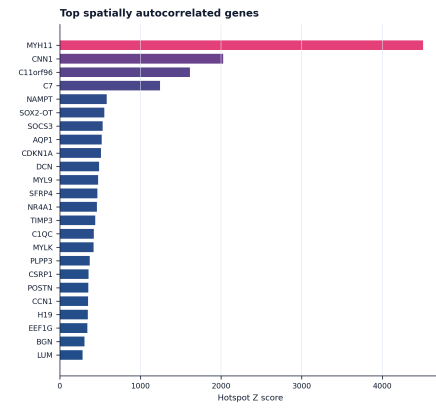
Supplementary Figure Legends

singleCellHaystack spatial non-randomness audit

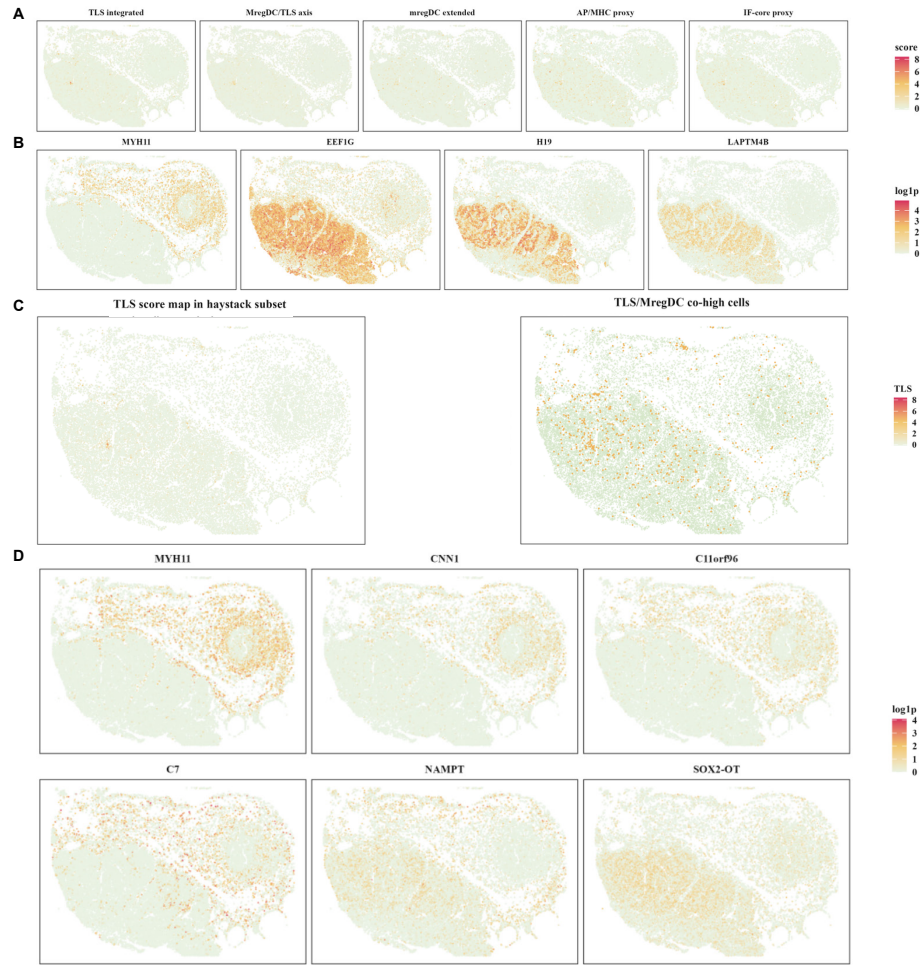


Supplementary Figure S1 singleCellHaystack spatial non-randomness audit. **A**, Top spatially non-random genes ranked by singleCellHaystack evidence in the deterministic ovarian cancer subset. **B**, Marker-level audit for TLS/mregDC/AP-associated genes, grouped by AP/MHC-II proxy, mregDC/AP, and TLS lymphoid marker categories.

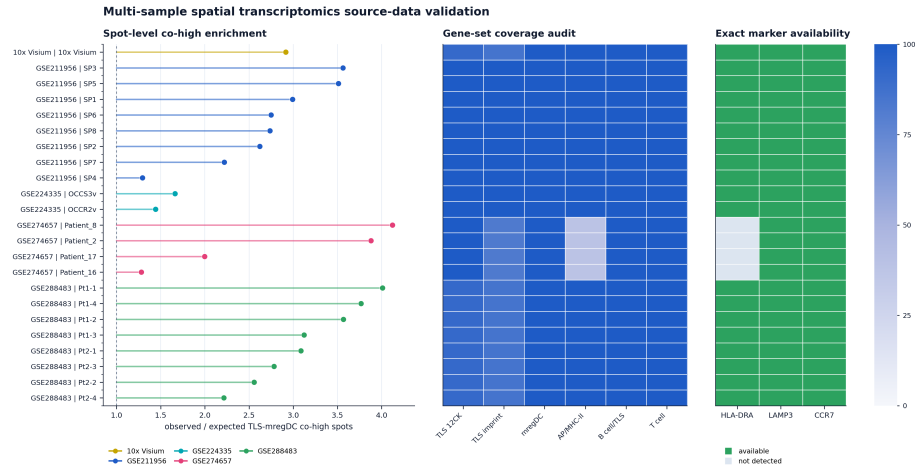
Hotspot spatial autocorrelation audit



Supplementary Figure S2 Hotspot spatial-autocorrelation audit. A, Top spatially autocorrelated genes ranked by Hotspot Z score in sampled Xenium cells. **B,** Marker-level Hotspot audit for TLS/mregDC/AP-associated genes, grouped by AP/MHC-II proxy, mregDC/AP, and TLS lymphoid marker categories.

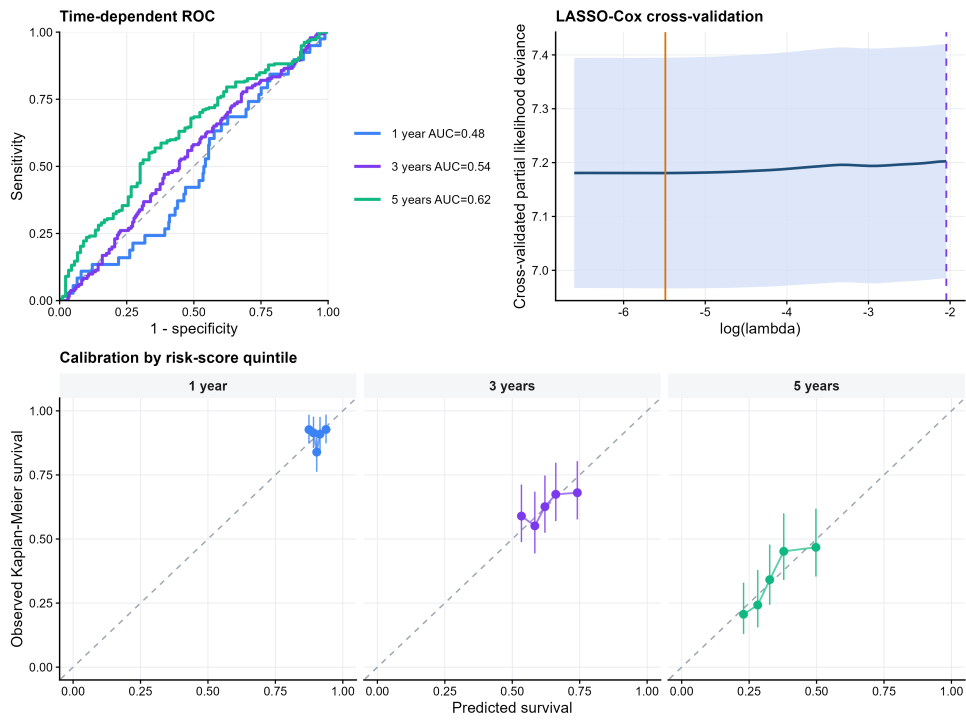


Supplementary Figure S3 Xenium spatial maps supporting feature audits. **A**, Spatial module maps for TLS-integrated, mregDC/TLS-axis, mregDC-extended, AP/MHC proxy, and IF-core proxy scores. **B**, Representative spatial expression maps for singleCellHaystack-prioritized non-random genes. **C**, TLS-score map and TLS/mregDC co-high cells in the same Xenium coordinate framework. **D**, Representative spatial expression maps for Hotspot-prioritized autocorrelated genes.

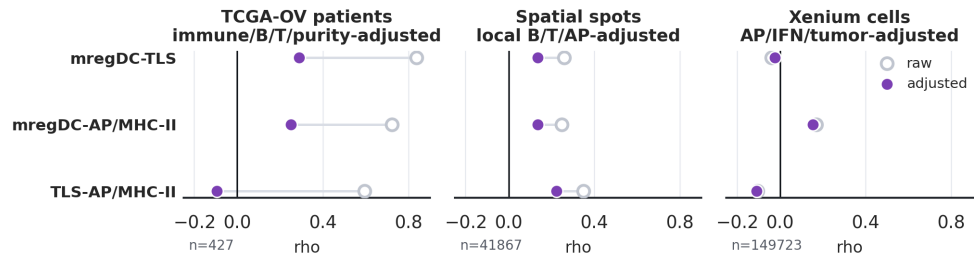


Supplementary Figure S4 Expanded multi-sample spatial source-data validation. **A**, Spot-level co-high enrichment across public ovarian spatial transcriptomic samples. **B**, Gene-set coverage audit for TLS, TLS-imprint, mregDC, MHC-II, IF-core, and T-cell-related signatures. **C**, Exact-marker availability for HLA-DRA, LAMP3, and CCR7 across the same datasets.

Exploratory prognostic model validation



Supplementary Figure S5 Exploratory model validation. **A**, Time-dependent ROC curves for 1-, 3-, and 5-year internal model assessment in TCGA-OV. **B**, LASSO-Cox cross-validation curve with selected penalty marker. **C**, Calibration-by-risk-quintile summaries for 1-, 3-, and 5-year predicted survival.



Supplementary Figure S6 Immune-richness-adjusted specificity audit. **A**, Raw and immune/B/T-cell/tumor-purity-adjusted Spearman correlations for mregDC-TLS, mregDC-AP/MHC-II, and TLS-AP/MHC-II associations in TCGA-OV patients. **B**, Raw and local B/T/AP-adjusted correlations for the same relationships across spatial transcriptomic spots. **C**, Raw and AP/IFN/tumor-adjusted correlations in Xenium cells.

Supplementary Source Data

- Current-numbering source-data map for main and supplementary figures: `source_data/SOURCE_DATA_INDEX.csv`.
- Fig. S1: `source_data/FigS01_singleCellHaystack_audit`.
- Fig. S2: `source_data/FigS02_Hotspot_audit`.
- Fig. S3: `source_data/FigS03_Xenium_spatial_maps`.
- Fig. S4: `source_data/FigS04_multi_sample_spatial_source`.
- Fig. S5: `source_data/FigS05_model_validation`.
- Fig. S6: `source_data/FigS06_specificity_adjustment`.
- Signature definitions, GEO source reconstruction, public spatial and 10x dataset accessions, and gene-coverage audits are provided in Supplementary Tables S1–S3 and `source_data/Tables_signature_and_GEO`.
- Legacy source-data folders are retained for traceability but should be interpreted through the current-numbering index.

Supplementary Tables

Supplementary Table S1 contains the reconstructed 22-sample scRNA-seq GEO source manifest, disease labels, sample counts, and accession-level summaries.

Supplementary Table S2 contains signature definitions, gene-coverage audits, marker-availability checks, and legacy-label harmonization for the mregDC, TLS, antigen-presentation/MHC-II, IFN, FDC/GC, HEV/stromal, proliferation, and EMT/invasion modules.

Supplementary Table S3 contains the 23 public multi-sample spatial transcriptomic samples used for Fig. 5/Fig. S4, including one public 10x Visium ovarian FFPE sample within the 23-sample spatial set, dataset-level GSE identifiers, sample-level GSM identifiers where applicable, and spot/cell counts. The same table also lists the separate public 10x scFFPE and Xenium Prime ovarian cancer reference datasets with URLs and software-version notes.